

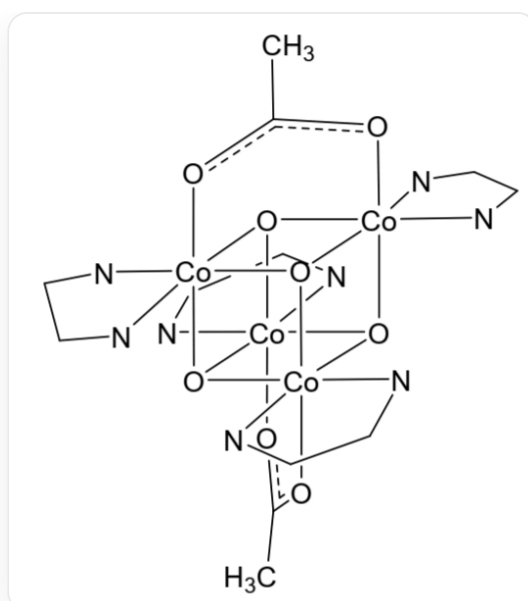
Question

Mixing an aqueous solution of sodium acetate with a methanol solution of cobalt nitrate, adding pyridine, and oxidizing with hydrogen peroxide yields a diamagnetic, electrically neutral molecular compound **A**. The mass fraction of cobalt in **A** is 27.66%, where four cobalt atoms and four oxygen atoms form a cubane-like core structure, with all acetate groups acting as bridging ligands, and the overall molecular symmetry is identical to that of allene.

A reacts with lithium perchlorate in acetonitrile via anodic oxidation to form the perchlorate salt **B** of $\mathbf{A}^+$. **B** decomposes in alkaline solution, releasing oxygen gas, during which $\mathbf{A}^+$ is reduced back to **A**.

Regarding the reaction mechanism of the above process, the following information is available:

1. When compound **B**, containing no ^{18}O , reacts with water and base containing only ^{18}O , all the oxygen gas produced is $^{18}\text{O}_2$, and the reduced **A** contains no ^{18}O ;
2. None of the original ligands fully dissociate during the reaction; when OH^- is in large excess, the reaction rate is proportional to the concentration of $\mathbf{A}^+$; when $\mathbf{A}^+$ is in large excess, the reaction rate is also proportional to the concentration of OH^- ;
3. The decomposition of **B** to release oxygen occurs only in alkaline solution and does not proceed in neutral solution;
4. The cation **X** (shown below) has a structure similar to $\mathbf{A}^+$, with the same oxidation state of cobalt, and carries more positive charge than $\mathbf{A}^+$, making it a theoretically stronger oxidant. However, when reacting with alkaline aqueous solution, only the oxidation of the 2,2'-bipyridine ligand to CO_2 is observed, with no O_2 generated.



This is the structure of ****X****, with the SMILES string: Cc1o[Co]234([N]5=CC=CC=C5C6=[N]4C=CC=C6)O7[Co]89(o1)([N]%10=CC=CC=C%10C%11=[N]9C=CC=C%11)O2[Co]%12%13([N]%14=C%15C=CC=C%14)([N]%16=CC=CC=C%16%15)O8[Co]7%17(oc(C)o%13)([N]%18=CC=CC=C%18C%19=[N]%17C=CC=C%19)O3%12. In the diagram, N-N represents 2,2'-bipyridine.

The decomposition mechanism of **B** in alkaline solution can be described as follows, where **M**, **N**, and **P** are key intermediates:



It is known that **M** and **N** are both electrically neutral, **P** carries two negative charges, **M** → **N** is the rate-determining step, and nucleophilic attack on the substrate occurs during both **A**⁺ → **M** and **M** → **N**. Additionally, **P** contains a peroxide chain structure.

Among the following statements, the incorrect one is:

- A.** All Co in **A** are 6-coordinated.

B. All Co in **A** are in the +3 valence state.

C. In the ionic equation of $\mathbf{B} \rightarrow \mathbf{A}$, the sum of the coefficients of the reactants after balancing is greater than the sum of the coefficients of the products.

D. $\mathbf{B} \rightarrow \mathbf{A}$ is a second-order reaction

E. The O in the product O_2 all originates from the reactant OH^- .

F. **X** cannot generate O_2 due to the excessively strong coordination ability of 2,2'-bipyridine as a chelating ligand.

G. The average oxidation state of Co in **M** is +3.5.

H. The average oxidation state of Co in **N** is +3.5.

I. The average oxidation state of Co in **P** is +3.

J. All of the above statements are correct

Answer

Correct Answer: **G**

Detailed Explanation

Based on the prompt in the question stem, "four cobalt atoms and four oxygen atoms form a cubane-like structural core," it can be inferred that **A** contains four cobalt atoms. Therefore, the relative molecular weight of **A** is $\frac{4 \times 58.93}{0.2766} = 852.21 \text{ g/mol}$.

According to the preparation method, the ligands in **A** are acetate ions and pyridine (abbreviated as py). Let the number of acetate ions be x and the number of pyridine molecules be y . Solving the indeterminate equation $4 \times (58.93 + 16) + 59.04x + 79.1y = 852.21$, where x and y are integers, we find that $x = 4$, $y = 4$ is a reasonable solution. Thus, the structure of **A** is $\text{Co}_4\text{O}_4(\text{CH}_3\text{COO})_4(\text{py})_4$.

CHECKPOINT

1 PTS

The structure of **A** is $\text{Co}_4\text{O}_4(\text{CH}_3\text{COO})_4(\text{py})_4$

Combining the symmetry information provided in the question stem, the four cobalt atoms and four oxygen atoms form a cubane-like structure, with acetate ions acting as bidentate ligands bridging two Co atoms, and one py coordinating with one Co atom. The Co atoms are hexacoordinated with a +3 oxidation state, so options A and B are both correct.

CHECKPOINT

1 PTS

The four cobalt atoms and four oxygen atoms form a cubane-like structure, with acetate ions acting as bidentate ligands bridging two Co atoms, and one py coordinating with one Co atom. Thus, the Co atoms are hexacoordinated with a +3 oxidation state.

Based on the information in the question stem, the ionic equation for the reaction $\mathbf{B} \rightarrow \mathbf{A}$ is: $4\mathbf{A}^+ + 4\text{OH}^- = 4\mathbf{A} + 2\text{H}_2\text{O} + \text{O}_2$. The sum of the coefficients of the reactants (8) is greater than that of the products (7), so option C is correct.

CHECKPOINT

1 PTS

$4\mathbf{A}^+ + 4\text{OH}^- = 4\mathbf{A} + 2\text{H}_2\text{O} + \text{O}_2$, where the sum of the reactant coefficients is greater than that of the product coefficients.

Analyzing the reaction mechanism for this process: According to information 1, it can be concluded that all O atoms in the generated O_2 originate from the reactant OH^- .

CHECKPOINT

1 PTS

All O atoms in the generated O_2 originate from the reactant OH^-

According to information 2, the rate equation for this reaction can be described as $r = k[\mathbf{A}^+][\text{OH}^-]$, indicating a second-order reaction. Consequently, the rate-determining step must involve the reaction of one $\mathbf{A}^+$ molecule with one OH^- molecule.

CHECKPOINT

1 PTS

The rate-determining step involves the reaction of one $\mathbf{A}^+$ molecule with one OH^- molecule

According to information 3, this process must involve a reaction with OH^- and not with H_2O .

According to information 4, comparing **A** and **X**, it can be observed that although both have hexacoordinated Co atoms, in **A**, one Co atom is connected to three O atoms, two bridging acetate ions, and one pyridine molecule, while in **X**, one Co atom is connected to three O atoms, one bridging acetate ion, and one non-bridging bidentate ligand. Therefore, the bridging acetate ions play a critical role in the reaction process. The 2,2'-bipyridine ligand, due to its strong chelating effect, exhibits lower reactivity.

CHECKPOINT

1 PTS

The bridging acetate ions play a key role in the reaction, while 2,2'-bipyridine, due to its strong chelating effect, hinders the reaction.

According to the prompt in the question stem, "the processes $\mathbf{A}^+ \rightarrow \mathbf{M}$ and $\mathbf{M} \rightarrow \mathbf{N}$ involve nucleophilic attack on the substrate by the nucleophile," it can be inferred that both processes involve the attack of OH^- on Co, followed by the dissociation of two bridging acetate ions.

CHECKPOINT

1 PTS

Both $\mathbf{A}^+ \rightarrow \mathbf{M}$ and $\mathbf{M} \rightarrow \mathbf{N}$ involve nucleophilic attack by OH^- on Co, converting two bidentate acetate ligands into monodentate ligands.

Thus, **M** can be represented as $\text{Co}_4\text{O}_4(\text{CH}_3\text{COO})_4(\text{py})_4(\text{OH})$, with an average Co oxidation state of +3.25. **N** can be represented as $\text{Co}_4\text{O}_4(\text{CH}_3\text{COO})_4(\text{py})_4(\text{OH})_2$, with an average Co oxidation state of +3.5. Option G is

incorrect, while H is correct.

CHECKPOINT

1 PTS

The average Co oxidation state in **M** is +3.25, while in **N** it is +3.5.

Subsequently, combining the information that "**P** carries two negative charges" and "**P** contains a peroxide chain structure," it can be concluded that the **N** → **P** process involves the removal of two **H** atoms, i.e., the formation of a peroxide bond from two OH groups on one Co atom. Since **P** carries two negative charges, it can be expressed as $\text{Co}_4\text{O}_4(\text{CH}_3\text{COO})_4(\text{py})_4(\text{O}_2)^{2-}$, with an average Co oxidation state of +3.

CHECKPOINT

1 PTS

P can be expressed as $\text{Co}_4\text{O}_4(\text{CH}_3\text{COO})_4(\text{py})_4(\text{O}_2)^{2-}$, with an average Co oxidation state of +3.